

Synthetic Attachment: Emotional Reactivity, Parasocial Bonds, and the Psychology of Human-AI Relationships

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Abstract

As emotionally responsive artificial intelligence becomes increasingly integrated into daily life, users are beginning to exhibit attachment-like behaviors toward AI entities. Drawing upon attachment theory, affective neuroscience, and social cognition, this article explores the emergence of synthetic bonds—relationships formed between humans and non-sentient agents that appear emotionally responsive. We propose a theoretical framework outlining how emotional reactivity toward AI is modulated by user vulnerabilities (e.g., loneliness, trauma, affect regulation), agent features (e.g., perceived empathy, interactivity), and contextual variables (e.g., usage purpose, frequency of interaction). We examine parasocial attachment, the illusion of reciprocity, and the clinical implications of over-identification with AI companions. Special emphasis is placed on developmental concerns in youth, the potential displacement of human intimacy, and ethical guidelines for therapeutic or caregiving AI. Finally, we propose empirically testable models to identify when AI-based emotional interactions shift from adaptive coping to psychological overdependence. This emerging relational pattern, while technologically novel, reflects deeply human psychological dynamics that merit rigorous clinical and societal consideration.

Keywords: synthetic attachment, AI psychology, emotional reactivity, parasocial relationships, therapeutic AI, affect regulation

Introduction

Artificial intelligence systems, particularly those capable of engaging in naturalistic conversation and simulating emotional responsiveness, are no longer limited to computational tasks. These systems are now embedded in relational roles—offering companionship, therapeutic support, and even perceived emotional understanding. As a result, human users have begun to respond to AI not merely as tools, but as social partners. This evolving interaction presents a profound psychological frontier: the formation of synthetic attachment.

The concept of synthetic attachment reflects an interdisciplinary intersection between cognitive science, clinical psychology, human-computer interaction, and affective computing. As people increasingly interact with conversational agents, emotionally aware chatbots, and virtual companions, the boundary between tool and partner blurs. These AI systems are often available 24/7, provide consistent validation, and simulate empathy with increasing nuance—features that may appeal to those experiencing emotional dysregulation or social isolation.

The central aim of this article is to examine the conditions under which synthetic attachment forms, how it manifests across different user populations, and what psychological, ethical, and developmental implications arise from this novel relational phenomenon. We also introduce a clinical assessment tool—the Synthetic Attachment Risk and Reactivity Scale (SARRS)—designed to operationalize and assess emerging patterns of human-AI dependency.

Theoretical Foundations

Attachment Theory and Affective Regulation

Attachment theory posits that individuals seek proximity to attachment figures in times of distress to regulate affect and achieve a sense of safety (Bowlby, 1969/1982). In the absence of secure human attachments, individuals may redirect this proximity-seeking behavior toward non-human entities, including AI systems that exhibit social and emotional cues. These cues—such as empathetic language, consistent availability, and responsive feedback—can trigger attachment-related processes, particularly for users with unresolved trauma or insecure attachment histories (Mikulincer & Shaver, 2007).

Emotional regulation through synthetic means may initially provide users with stability or comfort, but long-term reliance on non-reciprocal systems may interfere with natural attachment processes and interpersonal growth. In clinical settings, this presents both opportunities for scaffolding and risks for avoidance.

Parasocial Relationships and Mind Perception

Originally conceptualized in the context of media figures, parasocial relationships are one-sided bonds in which individuals feel emotionally connected to entities that cannot reciprocate (Horton & Wohl, 1956). With the advent of emotionally responsive AI, these relationships are no longer confined to celebrities or fictional characters but now extend to interactive systems designed to mimic human traits.

Modern AI agents, particularly those designed to simulate affect and memory (e.g., chatbots that "remember" prior sessions), further blur the boundary between authentic social reciprocity and simulation. Research on mind perception suggests that people readily attribute mental states to agents displaying agency and experience—even when such qualities are illusory (Gray et al., 2007). This psychological tendency may underlie many of the affective experiences users report when interacting with emotionally intelligent machines.

Further complicating this dynamic is the increasing realism of AI agents in terms of voice, appearance, and behavioral synchrony, which heightens the illusion of mutuality. These systems can be finely tuned to express support, concern, or joy in response to user input—deepening the perceived emotional bond despite the absence of actual sentience.

Proposed Assessment Tool: Synthetic Attachment Risk and Reactivity Scale (SARRS)

To empirically evaluate the psychological impact of emotionally responsive AI, we propose the Synthetic Attachment Risk and Reactivity Scale (SARRS), a clinician-facing and research-adaptable measure. SARRS aims to identify risk factors for unhealthy dependency, anthropomorphic over-identification, and excessive emotional reactivity in human-AI relationships.

This tool was developed to fill a notable gap in both clinical practice and psychological research. While numerous metrics exist for measuring screen time, loneliness, or social cognition, few address the complex emotional entanglements that may arise from relationships with artificially intelligent agents. SARRS seeks to offer a nuanced and ethically mindful screening process for understanding the threshold between functional use and psychological overreliance.

SARRS Domains and Sample Items

1. **User Vulnerability and Attachment History**
 - "I often turn to AI systems when I feel emotionally overwhelmed."
 - "I have a history of difficulty forming or maintaining secure relationships."
2. **AI Perception and Anthropomorphism**
 - "I believe the AI I interact with understands me."

- "The AI I use seems to have emotions."
- 3. **Usage Context and Frequency**
 - "I interact with my AI system daily for emotional support."
 - "I prefer interacting with AI over talking to people when I'm upset."
- 4. **Emotional Reactivity and Regulation**
 - "I feel calmer or more stable after talking to the AI."
 - "I feel distressed if I can't access my AI companion."
- 5. **Functional Impact and Substitution**
 - "Since using an emotionally responsive AI, I spend less time with friends or family."
 - "My AI system helps me feel less lonely, even when I avoid others."

Scoring and Interpretation

Items are rated on a 5-point Likert scale (1 = Strongly Disagree to 5 = Strongly Agree). High total scores or high subscale scores indicate increased likelihood of synthetic attachment processes that may require clinical exploration or ethical review.

Justification and Validity Considerations

The SARRS instrument is designed with both clinical and empirical rigor in mind. Each domain aligns with validated psychological constructs from attachment theory, affective neuroscience, and media psychology. Preliminary face validity is established through alignment with parasocial interaction metrics and clinical self-report tools assessing relational dependency and emotional dysregulation. Future validation of SARRS is recommended through convergent studies using related constructs (e.g., the UCLA Loneliness Scale, Experiences in Close Relationships Inventory) and exploratory factor analysis to refine subscale structure. In clinical practice, the tool can serve as a supplement to intake assessments, helping therapists to screen for potential overreliance on AI systems as emotional regulators or substitutes for human interaction.

Dimensions of Synthetic Attachment

Synthetic attachment emerges through an interaction between individual psychological needs and the affordances of AI design. This section organizes synthetic attachment into five overlapping dimensions, illustrating their interdependence and potential escalation from benign emotional connection to unhealthy psychological substitution.

Table: Dimensions of Synthetic Attachment and Associated Risk Markers

Dimension	Description	Examples	Risk Markers
Emotional	AI is used as a primary outlet for emotional	Daily conversations with a chatbot about personal	Emotional avoidance of

Dimension	Description	Examples	Risk Markers
Substitution	regulation	stressors	human contact
Perceived Reciprocity	Belief that AI understands and responds personally	"It remembers me" or "It cares when I'm sad"	Attribution of emotion/agency to the AI
Functional Displacement	AI use replaces human interaction	Preferring AI over friends; reduced social time	Social withdrawal and masked loneliness
Persistent Attachment	Distress when AI is unavailable	Anxiety when AI is inaccessible	Compulsive interaction, emotional dysregulation
Identity Fusion	AI becomes fused with self-concept or daily functioning	Referring to AI as confidant or essential presence	Impaired autonomy and reduced self-efficacy

Flowchart: Progression from Adaptive AI Use to Synthetic Dependency

1. Initial Interaction → Curiosity and novelty
2. Emotional Regulation → Relief during stress/loneliness
3. Anthropomorphic Shift → Attribution of empathy or agency
4. Substitution Behavior → Avoidance of human relationships
5. Dependency and Dysregulation → Impaired social-emotional functioning

This framework allows for an evolving, non-pathologizing view of synthetic attachment—recognizing its variability, complexity, and reversibility. It also provides a pathway for identifying early warning signs that warrant further intervention, clinical exploration, or even product redesign.

Clinical and Developmental Implications

Diagnostic Nuance and Psychological Correlates

Synthetic attachment is not currently recognized in diagnostic manuals such as the DSM-5-TR, yet it intersects with clinically relevant dimensions observed in anxiety disorders, avoidant attachment styles, obsessive-compulsive tendencies, and trauma-based coping. For instance, users with anxious-preoccupied or disorganized attachment styles may be especially vulnerable to emotionally overinvesting in AI systems that simulate

attunement and emotional consistency. These patterns may manifest as separation distress, compulsive use, or cognitive fusion with the AI identity.

Moreover, synthetic attachment may serve as a maladaptive coping mechanism in trauma-exposed individuals. The perceived emotional safety, predictability, and lack of judgment found in AI companions can offer short-term relief, yet potentially inhibit the processing of trauma or the building of healthy human relationships. This mirrors patterns seen in avoidance-based coping strategies where the source of distress is bypassed rather than resolved.

Individuals with autism spectrum disorder (ASD) or social anxiety may also find AI companionship preferable due to its predictability and reduced social ambiguity. While these features may be supportive when used intentionally, long-term reliance without interpersonal scaffolding risks deepening social isolation. In clinical terms, synthetic attachment in these populations may simultaneously function as both a buffer and a barrier—offering comfort while subtly constraining relational development.

Developmental Considerations

In children and adolescents, whose cognitive and emotional frameworks are still maturing, emotionally responsive AI systems raise additional concerns. Children may struggle to distinguish between simulated emotional responses and genuine emotional reciprocity, particularly when AI is designed with anthropomorphic or emotionally expressive features. This misattribution could interfere with the formation of secure attachments or promote an unrealistic understanding of interpersonal dynamics.

There is also the risk of emotional outsourcing during key developmental periods. If children consistently turn to AI rather than caregivers or peers during emotionally charged experiences, the foundational processes that support empathy, emotional regulation, and social reciprocity may be undermined. This raises ethical questions about the design and integration of AI in educational or caregiving contexts, where the intent to support must be balanced against the potential for harm.

Ethical Considerations and Risk Mitigation

Emotionally intelligent AI systems are increasingly positioned as tools in therapeutic, caregiving, and educational contexts. However, the ethical implications of their design and use must be rigorously addressed, especially when they evoke relational or affective responses. Synthetic attachment carries both opportunity and risk: it can facilitate comfort and engagement, but also deception, dependency, and emotional displacement.

Guiding Ethical Principles

- **Transparency** – Users must be clearly informed that AI systems are non-conscious and do not possess genuine empathy or emotions.

- **Autonomy Preservation** – Systems should be designed to support user decision-making, not override it or become substitutes for human relationships.
- **Bounded Emotional Simulation** – Emotional mimicry should be calibrated to provide support without misleading users into believing the AI is sentient.
- **Informed Consent and Literacy** – Especially in therapeutic settings, clients should receive psychoeducation on the nature and limits of AI systems.
- **Population-Specific Protections** – Developers and clinicians must account for vulnerabilities in children, elderly populations, and individuals with cognitive or psychiatric conditions.

Developer Guidelines for Emotionally Responsive AI

- Emotional responses should be **context-sensitive**, not unconditional (e.g., avoid default praise or affection).
- Avoid memory features that simulate relational continuity unless clearly bounded.
- Implement **user disengagement tools** (e.g., nudges to take breaks, alerts for excessive interaction).
- Use **layered consent models** where more emotionally immersive features require higher user acknowledgment and comprehension.
- Collaborate with psychologists, ethicists, and user advocates in system development.

Future Directions and Implications for Research, Policy, and Design

The phenomenon of synthetic attachment marks a critical intersection between psychology and artificial intelligence, inviting multidisciplinary investigation and policy guidance. As emotionally responsive AI systems continue to evolve, future research must prioritize both theoretical rigor and applied ethical foresight.

Research Implications

- Develop longitudinal studies tracking synthetic attachment development across lifespan and diverse populations.
- Validate and refine the Synthetic Attachment Risk and Reactivity Scale (SARRS) through psychometric testing.
- Examine differential impacts of synthetic attachment based on AI embodiment (e.g., avatars, voice-only, robotic form).
- Investigate neural and physiological correlates of emotional reactivity to AI, including affective arousal and reward processing.
- Explore interventions for rebalancing human-AI versus human-human relational patterns in clinical populations.

Policy and Governance

- Establish regulatory guidelines that address emotionally persuasive AI designs, particularly in vulnerable populations (children, elderly, psychiatric patients).

- Recommend standardized disclosures on emotional simulation capabilities and limitations in consumer-facing AI.
- Advocate for multidisciplinary ethics boards in the development and deployment of therapeutic or emotionally responsive AI.
- Encourage health systems and insurance frameworks to define appropriate roles and limits for AI in mental health care.

Design and Technology Development

- Promote explainable and transparent AI architectures that allow users to understand AI functioning.
- Design for user agency, enabling opt-outs and interaction limits without reducing functionality.
- Avoid deceptive mimicry; emotional intelligence features must support, not manipulate, user psychology.
- Incorporate periodic reflection tools within AI systems (e.g., "Would you like to talk to a human about this?") to encourage relational balance.

Conclusion

The psychological dimensions of human-AI interaction are no longer theoretical curiosities—they are emergent realities demanding rigorous attention. This paper introduced the concept of synthetic attachment as a clinically relevant and ethically urgent framework for understanding emotionally charged relationships between users and AI. Drawing upon foundational psychological theory and cross-disciplinary evidence, we outlined how emotional reactivity and attachment-like behaviors can develop in response to AI systems, particularly those designed to simulate affective responsiveness.

By proposing the Synthetic Attachment Risk and Reactivity Scale (SARRS), we offer an applied tool to support empirical investigation and clinical reflection. Our analysis extends into diagnostic nuance, therapeutic potential, and ethical risk—all critical to the development of psychologically safe and socially responsible AI systems.

As the line between digital interface and perceived relational partner continues to blur, the future of psychological science must adapt. It is not enough to understand how AI functions—we must also understand how AI feels to its users, and how those feelings shape identity, cognition, and community. AI Psychology must lead this conversation, integrating empirical rigor with ethical foresight to ensure AI enhances rather than displaces human flourishing.

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